

Summary and Recommendations

Executive Summary – Customer Churn Analysis

This analysis evaluates customer churn behavior across a dataset of **7,043 telecom customers**, with the objective of identifying key risk drivers and informing data-driven retention strategies.

The overall churn rate stands at **26.54%**, meaning approximately **1 in every 4 customers leaves the company**, while **73.46% are retained**. This level of churn is significant and indicates a need for targeted intervention, particularly within high-risk customer segments.

Key Drivers of Churn

1. Customer Tenure (Lifecycle Risk)

- Customers with **low tenure (0–12 months)** exhibit the **highest churn rates**, forming the majority of churned users.
- In contrast, customers with **long tenure (24+ months)** show significantly lower churn.
- This highlights a clear pattern: **churn risk declines as customer tenure increases**, making the **first year critical for retention efforts**.

2. Contract Type (Commitment Level)

- **Month-to-month contracts** have the **highest churn rate (~40–45%)**, making them the most vulnerable group.
- Customers on **one-year contracts** show moderate churn (~10–15%).
- Customers on **two-year contracts** have the **lowest churn (~2–5%)**.
- This demonstrates that **long-term contractual commitment reduces churn risk by up to 80–90% compared to month-to-month users**.

3. Service Adoption (Value Perception)

- Customers **without add-on services** (Online Security, Tech Support, Backup) churn at significantly higher rates.
- Customers who subscribe to **multiple services show reduced churn (often below 15–20%)**.

- This indicates that **service bundling increases perceived value and customer stickiness**.

4. Internet Service Type

- Customers using **Fiber Optic Internet** exhibit **higher churn rates (~40%+)** compared to DSL users (~15–20%).
- This suggests potential issues such as:
 - Higher pricing sensitivity
 - Service quality concerns
 - Mismatch between expectations and delivery

5. Demographic Factors

- **Senior Citizens** show a **higher churn rate (~40%)** compared to non-senior customers (~23%).
 - **Gender** does not significantly impact churn (male and female churn rates are nearly equal at ~26–27%).
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Behavioral Pattern Summary

High churn probability is strongly associated with:

- **Short tenure (<12 months)**
 - **Month-to-month contracts**
 - **Limited or no add-on services**
 - **Fiber optic internet usage**
 - **Senior citizen segment**
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Business Impact

- Losing **26.54% of customers** represents a substantial revenue leakage risk.
- Since acquisition costs are typically higher than retention costs, reducing churn even by **5–10% could significantly improve profitability and customer lifetime value (CLV)**.

Strategic Recommendations

To mitigate churn effectively, the company should focus on:

- **Early Lifecycle Engagement**
 - Strengthen onboarding and first-year experience to reduce early-stage churn.
- **Contract Conversion Strategy**
 - Incentivize migration from month-to-month plans to **long-term contracts**, potentially reducing churn by up to **80%**.
- **Service Bundling & Upselling**
 - Promote adoption of **value-added services**, increasing retention likelihood by **15–25%**.
- **Fiber Optic Service Optimization**
 - Investigate and address **pricing, quality, or expectation gaps** for fiber users.
- **Segment-Specific Retention**
 - Design targeted support and simplified service offerings for **senior customers**, a segment with ~40% churn risk.

Conclusion

Customer churn is primarily driven by **low engagement, weak contractual commitment, and insufficient service integration**. By targeting high-risk segments—particularly **new customers, month-to-month users, and low-service adopters**—the company can reduce churn, improve retention rates, and drive sustainable revenue growth.